Taylor's formulas and its applications

F18CE Week 3

1. Taylor's formula in one variable

Let $f : \mathbb{R} \to \mathbb{R}$ be an analytic function. Then we can write

$$f(x) = \sum_{r=0}^{n-1} \frac{f^{(r)}(a)}{r!}(x-a)^r + R_n(x),$$

where $p_{n-1}(x) := \sum_{r=0}^{n-1} \frac{f^{(r)}(a)}{r!}(x-a)^r$ is the Taylor's polynomial of degree $n-1$ centred at a, and $R_n(x) = \frac{(x-a)^n}{n!} f^{(n)}(\xi)$ for some ξ in between a and x, i.e., $\xi = a + c(x-a)$ for some $c \in [0,1]$.

If we know that $\left|f^{(n)}(\xi)\right| \leq M$, then $\left|R_n(x)\right| \leq \frac{\left|x-a\right|^n}{n!} M$.

2. Taylor's formula in two variables (Theorem 4.3)

If $f:\mathbb{R}^2 \to \mathbb{R}$ such that $\dfrac{\partial^n f}{\partial x^n}, \dfrac{\partial^n f}{\partial y^n}, \dfrac{\partial^n f}{\partial x^r \partial y^{n-r}}$ exist for $n = 1, 2, 3, \ldots$ at a point (x_0, y_0).

Let $\Delta x = x - x_0$ and $\Delta y = y - y_0$. Then

$$f(x,y) = f(x_0,y_0) + \sum_{r=1}^{\infty} \frac{1}{r!} \left(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y} \right)^r f(x_0,y_0),$$

$$\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)^1 f(x_0,y_0)=\Delta x\frac{\partial f(x_0,y_0)}{\partial x}+\Delta y\frac{\partial f(x_0,y_0)}{\partial y}=\Delta x f_x(x_0,y_0)+\Delta y f_y(x_0,y_0),$$

$$\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)^2 f(x_0,y_0)=(\Delta x)^2 f_{xx}(x_0,y_0)+2(\Delta x)(\Delta y)f_{xy}(x_0,y_0)+(\Delta y)^2 f_{yy}(x_0,y_0),$$

$$\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)^3 f(x_0,y_0)=(\Delta x)^3 f_{xxx}(x_0,y_0)+3(\Delta x)^2(\Delta y)f_{xxy}(x_0,y_0)$$
$$+3(\Delta x)(\Delta y)^2 f_{xyy}(x_0,y_0)+(\Delta y)^3 f_{yyy}(x_0,y_0),$$

etc...

(It is 'similar' to binomial expansion where we add on $f_{x\cdots xy\cdots y}$ depending on how many time of Δx and Δy occur.)

Taylor's formula in two variables with degree *n*

n	Taylor's formula for two variables with degree n
1	$P_1(x,y) = f(x_0,y_0) + \Delta x f_x(x_0,y_0) + \Delta y f_y(x_0,y_0)$
2	$P_2(x,y) = P_1(x,y) + \frac{1}{2}\left[(\Delta x)^2 f_{xx}(x_0,y_0) + 2(\Delta x)(\Delta y) f_{xy}(x_0,y_0) + (\Delta y)^2 f_{yy}(x_0,y_0)\right]$
3	$P_3(x,y) = P_2(x,y) + \frac{1}{3!}\left[(\Delta x)^3 f_{xxx}(x_0,y_0) + 3(\Delta x)^2(\Delta y) f_{xxy}(x_0,y_0) + 3(\Delta x)(\Delta y)^2 f_{xyy}(x_0,y_0) + (\Delta y)^3 f_{yyy}(x_0,y_0)\right]$
4	$P_4(x,y) = P_3(x,y) + \frac{1}{4!}\left[(\Delta x)^4 f_{xxxx}(x_0,y_0) + 4(\Delta x)^3(\Delta y) f_{xxxy}(x_0,y_0) + 6(\Delta x)^2(\Delta y)^2 f_{xxyy}(x_0,y_0) + 4(\Delta x)(\Delta y)^3 f_{xyyy}(x_0,y_0) + (\Delta y)^4 f_{yyyy}(x_0,y_0)\right]$

3. Remainder

We could write

$$f(x,y) = f(x_0,y_0) + \sum_{r=1}^{n} \frac{1}{r!}\left(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y}\right)^r f(x_0,y_0) + R_{n+1}(x_0,y_0)\,,$$

where

$$R_{n+1}(x,y) = \frac{1}{(n+1)!}\left(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y}\right)^{n+1} f(x+\theta\Delta x, y+\theta\Delta y)\,,$$

for some constant $0 \le \theta \le 1$.

4. Example 4.4

Consider the function $f(x,y) = \ln(x + y^2)$ at $(x_0, y_0) = (1,1)$.

Find the Taylor's formula of degree 1 and degree 2.

Solution. $f = \ln(x + y^2) \Rightarrow f(1,1) = \ln 2$

$$f_x = \frac{1}{x+y^2} \quad \Rightarrow f_x(1,1) = \frac{1}{2}$$

$$f_y = \frac{2y}{x+y^2} \quad \Rightarrow f_y(1,1) = 1$$

$$f_{xx} = \frac{-1}{(x+y^2)^2} \quad \Rightarrow f_{xx}(1,1) = -\frac{1}{4}$$

$$f_{yy} = \frac{2(x-y^2)}{(x+y^2)^2} \quad \Rightarrow f_{yy}(1,1) = 0$$

$$f_{xy} = \frac{-2y}{(x+y^2)^2} \quad \Rightarrow f_{xy}(1,1) = -\frac{1}{2}$$

Answer for Linear approximation

$$\Delta x = x - x_0 = x - 1\,,\ \Delta y = y - y_0 = y - 1$$

$$P_1(x,y) = f(x_0,y_0) + \Delta x f_x(x_0,y_0) + \Delta y f_y(x_0,y_0)$$

$$P_1(x,y) = f(1,1) + (x-1)f_x(1,1) + (y-1)f_y(1,1)$$

$$= \ln 2 + \frac{1}{2}(x-1) + y - 1$$

Answer for Quadratic approximation

$$\Delta x = x - x_0 = x - 1\,,\ \Delta y = y - y_0 = y - 1$$

$$P_2(x,y) = P_1(x,y) + \frac{1}{2}\Big[(\Delta x)^2 f_{xx}(x_0,y_0) + 2(\Delta x)(\Delta y) f_{xy}(x_0,y_0) + (\Delta y)^2 f_{yy}(x_0,y_0)\Big]$$

$$P_2(x,y) = \ln 2 + \frac{1}{2}(x-1) + (y-1) - \frac{1}{8}(x-1)^2 - \frac{1}{2}(x-1)(y-1)$$

5. Linear approximation

Formula for linear approximation of two variables (derived from Taylors expansion)

$$f(x,y) \approx f(x_0,y_0) + (x-x_0)f_x(x_0,y_0) + (y-y_0)f_y(x_0,y_0)\,. \quad (1)$$

We see that this just says that the tangent plane to the surface $z = f(x,y)$ at the point.

Let $\Delta x = x - x_0,\ \Delta y = y - y_0$. Then we get rid of terms x and y.

$$f(\Delta x + x_0, \Delta y + y_0) \approx f(x_0, y_0) + \Delta x f_x(x_0, y_0) + \Delta y f_y(x_0, y_0). \quad (2)$$

Let df or $\Delta f = f(\Delta x + x_0, \Delta y + y_0) - f(x_0, y_0)$. Then we can further simplify (2) into

$$df = \Delta f \approx \Delta x f_x(x_0, y_0) + \Delta y f_y(x_0, y_0). \quad (3)$$

df is called the differential of *f*, meaning the difference of f(x,y) and f(x_0 , y_0)

6. Quadratic approximation

Formula for quadratic approximation of two variables (derived from Taylors expansion)

$$f(x,y) \approx f(x_0,y_0) + (x-x_0)f_x(x_0,y_0) + (y-y_0)f_y(x_0,y_0) + \tfrac{1}{2}(x-x_0)^2 f_{xx}(x_0,y_0) + \tfrac{1}{2}(y-y_0)^2 f_{yy}(x_0,y_0) + (x-x_0)(y-y_0)f_{xy}(x_0,y_0). \quad (4)$$

We see that this just says that the tangent plane to the surface $z = f(x,y)$ at the point.

Let $\Delta x = x - x_0,\ \Delta y = y - y_0$. Then get rid of terms x and y, we have

$$\begin{aligned} f(\Delta x + x_0, \Delta y + y_0) \approx\ & f(x_0, y_0) + \Delta x f_x(x_0, y_0) + \Delta y f_y(x_0, y_0) \\ & + \tfrac{1}{2}(\Delta x)^2 f_{xx}(x_0, y_0) + \tfrac{1}{2}(\Delta y)^2 f_{yy}(x_0, y_0) + (\Delta x)(\Delta y) f_{xy}(x_0, y_0) \end{aligned} . \quad (5)$$

Let df or $\Delta f = f(\Delta x + x_0, \Delta y + y_0) - f(x_0, y_0)$. Then we can further simplify (5) into

$$\begin{aligned} df = \Delta f \approx\ & \Delta x f_x(x_0, y_0) + \Delta y f_y(x_0, y_0) + (\Delta x)(\Delta y) f_{xy}(x_0, y_0) . \quad (6) \\ & + \tfrac{1}{2}(\Delta x)^2 f_{xx}(x_0, y_0) + \tfrac{1}{2}(\Delta y)^2 f_{yy}(x_0, y_0) \end{aligned}$$

7. Example 4.2 (linear approximation)

Find the linear approximation for $\frac{0.998^3}{1.003}$.

Solution. Let $f(x,y)=\frac{x^3}{y}$. Then $f_x=\frac{3x^2}{y}$, $f_y=-\frac{x^3}{y^2}$.

Let $x=0.998$, $y=1.003$ and $x_0=1=y_0$.

Then by using $f(x,y)\approx f(x_0,y_0)+(x-x_0)f_x(x_0,y_0)+(y-y_0)f_y(x_0,y_0)$

$$\frac{0.998^3}{1.003}=f(0.998,1.003)\approx f(1,1)+(0.998-1)f_x(1,1)+(1.003-1)f_y(1,1)$$
$$=1-0.002(3)+0.003(-1)=0.991$$

Note that the exact answer is 0.9910389 (correct to seven decimal places).

8. Revisit Example 4.2 (quadratic approximation)

Find the quadratic approximate value for $\frac{0.998^3}{1.003}$.

Solution. From the last slide, we have the linear approximation $\frac{0.998^3}{1.003} \approx 0.991$

Now adding on $+\frac{1}{2}(\Delta x)^2 f_{xx}(x_0, y_0) + \frac{1}{2}(\Delta y)^2 f_{yy}(x_0, y_0) + (\Delta x)(\Delta y) f_{xy}(x_0, y_0)$,

$f(x,y) = \frac{x^3}{y} \Rightarrow f_x = \frac{3x^2}{y}$, $f_y = -\frac{x^3}{y^2} \Rightarrow f_{xx} = \frac{6x}{y}$, $f_{yy} = \frac{2x^3}{y^3}$, $f_{xy} = -\frac{3x^2}{y^2}$, and

$x = 0.998$, $y = 1.003$, $x_0 = 1 = y_0 \Rightarrow \Delta x = -0.002$, $\Delta y = 0.003$. We have

$$\frac{0.998^3}{1.003} \approx 0.991 + \frac{1}{2}(0.002^2)(6) + \frac{1}{2}(0.003^2)(2) + (-0.002)(0.003)(-3)$$
$$= 0.991 + 0.000039 = 0.991039$$

By way of comparison, the exact error is 0.9910389 (to seven decimal places).

9. General Taylor's formula for three variable functions

$$f(x,y,z) = f(x_0,y_0,z_0) + \sum_{k=1}^{\infty} \frac{1}{k!}(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y} + \Delta z \frac{\partial}{\partial z})^k f(x_0,y_0,z_0)$$

10. Taylor's formula for three variable functions with degree 1

$$P_1(x,y,z) = f(x_0,y_0,z_0) + (\Delta x)f_x(x_0,y_0,z_0) + (\Delta y)f_y(x_0,y_0,z_0) + (\Delta z)f_z(x_0,y_0,z_0)$$

Linear approximation: $f(x,y,z) \approx P_1(x,y,z)$

11. Taylor's formula for three variable functions with degree 2

$$P_2(x,y,z) = P_1(x,y,z) + \tfrac{1}{2}\Big((\Delta x)^2 f_{xx}(x_0,y_0,z_0) + (\Delta y)^2 f_{yy}(x_0,y_0,z_0) + (\Delta z)^2 f_{zz}(x_0,y_0,z_0)\Big)$$
$$+(\Delta x)(\Delta y) f_{xy}(x_0,y_0,z_0) + (\Delta x)(\Delta z) f_{xz}(x_0,y_0,z_0) + (\Delta y)(\Delta z) f_{yz}(x_0,y_0,z_0)$$

Quadratic approximation: $f(x,y,z) \approx P_2(x,y,z)$

12. Extra example (a) (linear approximation)

Linearly approximate $\sqrt{4.2} + \sqrt[3]{26.7} + \sqrt[4]{256.4}$.

Solution. Let $f(x, y, z) = x^{1/2} + y^{1/3} + z^{1/4}$.

Then $f_x = \frac{1}{2} x^{-1/2}$, $f_y = \frac{1}{3} y^{-2/3}$, $f_z = \frac{1}{4} z^{-3/4}$.

Let $x = 4.2,\ y = 26.7,\ z = 256.4$; and $x_0 = 4,\ y_0 = 27,\ z_0 = 256$. By using

$$f(x,y,z) \approx f(x_0,y_0,z_0) + (x-x_0)f_x(x_0,y_0,z_0) + (y-y_0)f_y(x_0,y_0,z_0) + (z-z_0)f_z(x_0,y_0,z_0)$$

$$\sqrt{4.2} + \sqrt[3]{26.7} + \sqrt[4]{256.4} \approx f(4,27,256) + 0.2f_x(4,27,256) - 0.3f_y(4,27,256) + 0.4f_z(4,27,2$$

$$= 2 + 3 + 4 + 0.2 \times \tfrac{1}{4} - 0.3 \times \tfrac{1}{27} + 0.4 \times \tfrac{1}{256} = 9.0405$$

Note that the exact answer is 9.03980 (correct to five decimal places), a difference of about 0.008%.

13. Revisit Extra example (a) (quadratic approximation)

Find the quadratic approximate value for $\sqrt{4.2}+\sqrt[3]{26.7}+\sqrt[4]{256.4}$.

Solution. $f_{xx}=-\frac{1}{4}x^{-3/2}, f_{yy}=-\frac{2}{9}y^{-5/3}, f_{zz}=-\frac{3}{16}z^{-7/4}, f_{xy}=0=f_{xz}=f_{yz}$

$$\tfrac{1}{2}\big(\Delta x\big)^2 f_{xx}(x_0,y_0,z_0)+\tfrac{1}{2}\big(\Delta y\big)^2 f_{yy}(x_0,y_0,z_0)+\tfrac{1}{2}\big(\Delta z\big)^2 f_{zz}(x_0,y_0,z_0)$$
$$+\big(\Delta x\big)\big(\Delta y\big) f_{xy}(x_0,y_0,z_0)+\big(\Delta x\big)\big(\Delta z\big) f_{xz}(x_0,y_0,z_0)+\big(\Delta y\big)\big(\Delta z\big) f_{yz}(x_0,y_0,z_0)$$
$$=\tfrac{1}{2}\big(0.2\big)^2\big(-\tfrac{1}{32}\big)+\tfrac{1}{2}\big(-0.3\big)^2\big(-\tfrac{2}{2187}\big)+\tfrac{1}{2}\big(0.4\big)^2\big(-\tfrac{3}{262144}\big)$$
$$=-0.0006670677907$$

Then the quadratic approximation is $9.040451389-0.0006670677907=9.039784321$.

The exact answer is 9.039799219 and our approximation is correct up to four decimal places.

14. Extra example (b) (linear approximation)

A triangle ABC has sides $BC = a = 10.1\text{cm}$ and $AC = b = 19.8\text{cm}$ which include an angle 35°. Approximate the area of the triangle.

Solution. The area of the triangle is given by $f(a,b,\theta) = \frac{1}{2}ab\sin\theta$, where θ is the angle (measuring in radian) between adjacent sides with length a and b.

Now $f_a(a,b,\theta) = \frac{1}{2}b\sin\theta$, $f_b(a,b,\theta) = \frac{1}{2}a\sin\theta$, $f_c(a,b,\theta) = \frac{1}{2}ab\cos\theta$.

Let $(a,b,\theta) = \left(10.1, 19.8, \frac{7\pi}{36}\right)$ and $(a_0,b_0,\theta_0) = \left(10, 20, \frac{\pi}{6}\right)$. Then

$(\Delta a, \Delta b, \Delta\theta) = \left(0.1, -0.2, \frac{\pi}{36}\right)$. By using

$$f(a,b,\theta) \approx f(a_0,b_0,\theta_0) + \Delta a f_x(a_0,b_0,\theta_0) + \Delta b f_y(a_0,b_0,\theta_0) + \Delta\theta f_z(a_0,b_0,\theta_0),$$

$$
\begin{aligned}
f\left(10.1, 19.8, \tfrac{7\pi}{36}\right) &\approx f\left(10, 20, \tfrac{\pi}{6}\right) + 0.1 f_x\left(10, 20, \tfrac{\pi}{6}\right) - 0.2 f_y\left(10, 20, \tfrac{\pi}{6}\right) + \tfrac{\pi}{36} f_z\left(10, 20, \tfrac{\pi}{6}\right) \\
&= \tfrac{10(20)}{2} \sin\tfrac{\pi}{6} + 0.1 \times \tfrac{20}{2} \sin\tfrac{\pi}{6} - 0.2 \times \tfrac{10}{2} \sin\tfrac{\pi}{6} - \tfrac{\pi}{72}(10 \times 20) \cos\tfrac{\pi}{6} \\
&= 50 + \tfrac{1}{2} - \tfrac{1}{2} + \tfrac{50\sqrt{3}\pi}{36} = 50\left(1 + \tfrac{\sqrt{3}\pi}{36}\right) \approx 57.56
\end{aligned}
$$

The exact answer is 57.35 to two decimal places, the approximation has an error of about 0.37%.

15. Remainder/Error term

$$R_1(x,y) = (x-x_0)f_x(\tilde{x},\tilde{y}) + (y-y_0)f_y(\tilde{x},\tilde{y})$$

$$R_2(x,y) = \frac{1}{2}\Big((x-x_0)^2 f_{xx}(\tilde{x},\tilde{y}) + (y-y_0)^2 f_{yy}(\tilde{x},\tilde{y}) + 2(x-x_0)(y-y_0)f_{xy}(\tilde{x},\tilde{y})\Big)$$

$$R_3(x,y) = \frac{1}{6}\Big((x-x_0)^3 f_{xxx}(\tilde{x},\tilde{y}) + (y-y_0)^3 f_{yyy}(\tilde{x},\tilde{y}) + 3(x-x_0)^2(y-y_0)f_{xxy}(\tilde{x},\tilde{y}) + 3(x-x_0)(y-y_0)^2 f_{xyy}(\tilde{x},\tilde{y})\Big)$$

where $\tilde{x}$ is in between x and x_0; $\tilde{y}$ is in between y and y_0,

$R_2(x,y)$ is the error term in the linear approximation, and

$R_3(x,y)$ is the error term in the quadratic approximation.

Example (a)

Find the error term in the linear approximation $0.99^2 + 1.1^2$.

$f(x,y) = x^2 + y^2, (x_0, y_0) = (1,1)$.

(Exact answer is 2.1901).

$f_x = 2x, f_y = 2y, f_{xx} = 2 = f_{yy}, f_{xy} = 0$

$f(1,1) = 2, f_x(1,1) = 2 = f_y(1,1)$

$$P_1(x,y) = 2 + 2(x-1) + 2(y-1) = 2x + 2y - 2$$

$$0.99^2 + 1.1^2 = f(0.99,1.1) \approx P_1(0.99,1.1) = 2(0.99) + 2(1.1) - 2 = 2.18\,.$$

$$\begin{aligned} R_2(x,y) &= \tfrac{1}{2}\Big((x-x_0)^2 f_{xx}(\tilde{x},\tilde{y}) + (y-y_0)^2 f_{yy}(\tilde{x},\tilde{y}) + 2(x-x_0)(y-y_0) f_{xy}(\tilde{x},\tilde{y})\Big) \\ &= \tfrac{1}{2}\Big(2(x-1)^2 + 2(y-1)^2\Big) \\ &= (x-1)^2 + (y-1)^2 \end{aligned}$$

So, $R_2(0.99,1.1) = (0.99-1)^2 + (1.1-1)^2 = 0.0101\,.$

Note that $2.18 + 0.0101 = 2.1901\,.$

Example (b)

Let $f(x,y) = 2e^{x+y}$.

Write down $R_2(x,y), R_3(x,y)$ for Taylors polynomial of f at (0,0).

Hence, deduce $R_n(x,y)$.

Solution. $f(x,y) = 2e^{x+y}$. Note that all partial derivatives have the same answer.

So, $f(0,0) = 2$ and so as all the other partial derivatives at (0,0). So,

$$
\begin{aligned}
P_1(x,y) &= 2 + 2x + 2y \\
P_2(x,y) &= 2 + 2x + 2y + x^2 + y^2 + 2xy \\
R_2(x,y) &= \frac{1}{2}\Big(x^2 f_{xx}(\tilde{x},\tilde{y}) + y^2 f_{yy}(\tilde{x},\tilde{y}) + 2xy f_{xy}(\tilde{x},\tilde{y})\Big) \\
&= \frac{1}{2}\Big(x^2 \times 2e^{\tilde{x}+\tilde{y}} + y^2 \times 2e^{\tilde{x}+\tilde{y}} + 2xy \times 2e^{\tilde{x}+\tilde{y}}\Big) \\
&= e^{\tilde{x}+\tilde{y}}(x+y)^2 \\
R_3(x,y) &= \frac{1}{3}e^{\tilde{x}+\tilde{y}}(x+y)^3 \\
R_n(x,y) &= \frac{2}{n!}e^{\tilde{x}+\tilde{y}}(x+y)^n
\end{aligned}
$$

16. Cubic one-variable vs quadratic two-variable vs linear three-variable

Suppose we approximate $0.99^2 + \ln 1.01 + e^{0.01}$ using three different functions:

(a) $f(x) = (1-x)^2 + \ln(1+x) + e^x$ at $x_0 = 0$ using the cubic approximation

(b) $g(x,y) = x^2 + \ln(1+y) + e^y$ at $(x_0, y_0) = (1,0)$ using the quadratic approximation

(c) $h(x,y,z) = x^2 + \ln y + e^z$ at $(x_0, y_0, z_0) = (1,1,0)$ using the linear approximation

Which answer is the most accurate one?

Solution. (a) $f(x) = (1-x)^2 + \ln(1+x) + e^x \quad \Rightarrow f(0) = 2$

$f'(x) = -2(1-x) + \frac{1}{1+x} + e^x \quad \Rightarrow f'(0) = 0$

$f''(x) = 2 - \frac{1}{(1+x)^2} + e^x \quad \Rightarrow f''(0) = 2$

$f'''(x) = \frac{2}{(1+x)^3} + e^x \quad \Rightarrow f'''(0) = 3$

Cubic approx.: $f(x) \approx 2 + \frac{2}{2!}x^2 + \frac{3}{3!}x^3 = 2 + x^2 + \frac{1}{2}x^3$

$\therefore f(0.01) \approx 2 + 0.01^2 + \frac{1}{2}(0.01)^3 = 2.0001005$

(exact answer: 2.000100498)

(b) $g(x,y) = x^2 + \ln(1+y) + e^y \Rightarrow g(1,0) = 2$

$g_x = 2x \Rightarrow g_x(1,0) = 2$

$g_y = \frac{1}{1+y} + e^y \Rightarrow g_y(1,0) = 2$

$g_{xx} = 2 \Rightarrow g_{xx}(1,0) = 2$

$g_{yy} = -\frac{1}{(1+y)^2} + e^y \Rightarrow g_{yy}(1,0) = 0$

$g_{xy} = 0 \Rightarrow g_{xy}(1,0) = 0$

Quadratic approx.:

$$g(x,y) \approx g(x_0,y_0) + (\Delta x) g_x(x_0,y_0) + (\Delta y) g_y(x_0,y_0)$$
$$+ \tfrac{1}{2}\Big((\Delta x)^2 g_{xx}(x_0,y_0) + 2(\Delta x)(\Delta y) g_{xy}(x_0,y_0) + (\Delta y)^2 g_y(x_0,y_0)\Big)$$

$$x = 0.99,\ \ y = 1.01,\ \ x_0 = 1,\ \ y_0 = 0,\ \ \Delta x = -0.01,\ \ \Delta y = 0.01.$$

$$g(x,y) \approx 2 + 2(x-1) + 2y + \tfrac{1}{2}(2(x-1)^2 + 0 + 0) = 2 + 2\Delta x + 2\Delta y + (\Delta x)^2$$

$$\therefore\ g(0.99,1.01) \approx 2 + 2(-0.01) + 2(0.01) + (-0.01)^2 = 2.0001$$

(c) $h = x^2 + \ln y + e^z \quad \Rightarrow h(1,1,0) = 2$

$h_x = 2x \quad \Rightarrow h_x(1,1,0) = 2$

$h_y = \frac{1}{y} \quad \Rightarrow h_y(1,1,0) = 1$

$h_z = e^z \quad \Rightarrow h_z(1,1,0) = 1$

Linear approx.: $h(x,y,z) \approx h(x_0,y_0,z_0) + (\Delta x)h_x(x_0,y_0,z_0)$

$+(\Delta y)h_y(x_0,y_0,z_0) + (\Delta z)h_z(x_0,y_0,z_0)$

$$x = 0.99,\ \ y = 1.01,\ \ z = 0.01,\ \ x_0 = 1,\ \ y_0 = 1,\ \ z_0 = 0,$$
$$\Delta x = -0.01,\ \ \Delta y = 0.01,\ \ \Delta z = 0.01.$$

$$h(x, y, z) \approx 2 - 0.01(2) + (0.01)(1) + (0.01)(1) = 2\,.$$

Note that

$$h(0.99, 1.01, 0.01) = 2$$
$$< g(0.99, 1.01) = 2.0001$$
$$< f(0.01) = 2.0001005$$

and exact answer is 2.000100498...

The ultimate Taylors' formula

Let $x = (x_1, x_2, \ldots, x_n)$ be an n-variable vector and $a = (a_1, a_2, \ldots, a_n)$ be a point.

Let $f(x) : \mathbb{R}^n \to \mathbb{R}$ be a function. Then the Taylors' formula of f at point a is

$$f(x) = f(a) + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\left(\sum_{i=1}^{n} \Delta x_i \frac{\partial}{\partial x_i} \right)^k f(a) \right).$$